

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk:

Product Description:

Cat No. :

Synonyms

CAS No

Molecular Formula

**Tetramethylammonium perchlorate**
**Tetramethylammonium perchlorate**

420560000; 420560250

Methanaminium, N,N,N-trimethyl-, perchlorate.

2537-36-2

C4 H12 Cl N O4

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
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Tel: +03-5525 7888

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 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                   |
|----------------------------------------------------|-------------------|
| Oxidizing solids                                   | Category 2 (H272) |
| Skin Corrosion/Irritation                          | Category 2 (H315) |
| Serious Eye Damage/Eye Irritation                  | Category 2 (H319) |
| Specific target organ toxicity - (single exposure) | Category 3 (H335) |

**Label Elements**


Signal Word

Danger

Hazard Statements

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H272 - May intensify fire; oxidizer  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation  
H315 - Causes skin irritation

## Precautionary Statements

### Prevention

P220 - Keep away from clothing and other combustible materials  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P221 - Take any precaution to avoid mixing with combustibles  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P271 - Use only outdoors or in a well-ventilated area  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                    | CAS No    | Weight % |
|----------------------------------------------|-----------|----------|
| Methanaminium, N,N,N-trimethyl-, perchlorate | 2537-36-2 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.                          |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Get medical attention.                                                                                                   |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration.                         |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### Most important symptoms and effects, both acute and delayed

No information available.

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## Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Oxidizer: Contact with combustible/organic material may cause fire. Containers may explode when heated. May ignite combustibles (wood, paper, oil, clothing, etc.).

### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Hydrogen chloride gas.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Keep combustibles (wood, paper, oil, etc) away from spilled material. Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Do not breathe dust. Do not get in eyes, on skin, or on clothing. Keep away from clothing and other combustible materials.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Do not store near combustible materials.

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## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

#### Exposure Controls

##### **Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

##### **Personal protective equipment**

###### **Eye Protection**

Goggles

###### **Hand Protection**

Protective gloves

###### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

###### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

###### **Recommended Filter type:**

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### **Appearance**

White

#### **Physical State**

Powder Solid

#### **Odor**

No information available

#### **Odor Threshold**

No data available

#### **pH**

No information available

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|                     |                          |                                   |
|---------------------|--------------------------|-----------------------------------|
| Melting Point/Range | 300 °C / 572 °F          |                                   |
| Softening Point     | No data available        |                                   |
| Boiling Point/Range | No information available |                                   |
| Flash Point         | No information available | Method - No information available |

|                          |                          |       |
|--------------------------|--------------------------|-------|
| Evaporation Rate         | Not applicable           | Solid |
| Flammability (solid,gas) | No information available |       |
| Explosion Limits         | No data available        |       |

|                              |                          |       |
|------------------------------|--------------------------|-------|
| Vapor Pressure               | No data available        |       |
| Vapor Density                | Not applicable           | Solid |
| Specific Gravity / Density   | No data available        |       |
| Bulk Density                 | No data available        |       |
| Water Solubility             | No information available |       |
| Solubility in other solvents | No information available |       |

## Partition Coefficient (n-octanol/water)

|                           |                          |       |
|---------------------------|--------------------------|-------|
| Autoignition Temperature  | No data available        |       |
| Decomposition Temperature | No data available        |       |
| Viscosity                 | Not applicable           | Solid |
| Explosive Properties      | No information available |       |
| Oxidizing Properties      | Oxidizer                 |       |

|                   |                |
|-------------------|----------------|
| Molecular Formula | C4 H12 Cl N O4 |
| Molecular Weight  | 173.6          |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

Yes.

### Chemical Stability

Oxidizer: Contact with combustible/organic material may cause fire. Heating may cause an explosion.

### Possibility of Hazardous Reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
| Hazardous Reactions      | No information available. |

### Conditions to Avoid

Avoid dust formation. Incompatible products. Excess heat. Combustible material.

### Incompatible Materials

Reducing Agent. Acids. Finely powdered metals. Organic materials. Strong reducing agents. Combustible material.

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## Hazardous Decomposition Products

Nitrogen oxides (NOx). Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

|                                                   |                                                                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product Information</b>                        | No acute toxicity information is available for this product                                                                         |
| <b>(a) acute toxicity;</b>                        |                                                                                                                                     |
| <b>Oral</b>                                       | No data available                                                                                                                   |
| <b>Dermal</b>                                     | No data available                                                                                                                   |
| <b>Inhalation</b>                                 | No data available                                                                                                                   |
| <b>(b) skin corrosion/irritation;</b>             | Category 2                                                                                                                          |
| <b>(c) serious eye damage/irritation;</b>         | Category 2                                                                                                                          |
| <b>(d) respiratory or skin sensitization;</b>     |                                                                                                                                     |
| <b>Respiratory</b>                                | No data available                                                                                                                   |
| <b>Skin</b>                                       | No data available                                                                                                                   |
| <b>(e) germ cell mutagenicity;</b>                | No data available                                                                                                                   |
| <b>(f) carcinogenicity;</b>                       | No data available                                                                                                                   |
|                                                   | There are no known carcinogenic chemicals in this product                                                                           |
| <b>(g) reproductive toxicity;</b>                 | No data available                                                                                                                   |
| <b>(h) STOT-single exposure;</b>                  | Category 3                                                                                                                          |
| <b>Results / Target organs</b>                    | Respiratory system.                                                                                                                 |
| <b>(i) STOT-repeated exposure;</b>                | No data available                                                                                                                   |
| <b>Target Organs</b>                              | No information available.                                                                                                           |
| <b>(j) aspiration hazard;</b>                     | Not applicable                                                                                                                      |
|                                                   | Solid                                                                                                                               |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                      |
| <b>Symptoms / effects, both acute and delayed</b> | No information available.                                                                                                           |
| <b>Endocrine Disrupting Properties</b>            | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |

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## SECTION 12: ECOLOGICAL INFORMATION

|                                        |                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <u>Ecotoxicity effects</u>             | Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. |
| <u>Persistence and degradability</u>   | No information available                                                                                                    |
| <u>Bioaccumulative potential</u>       | No information available                                                                                                    |
| <u>Mobility in soil</u>                | No information available.                                                                                                   |
| <u>Endocrine Disruptor Information</u> | This product does not contain any known or suspected endocrine disruptors                                                   |
| <u>Other adverse effects</u>           | No information available                                                                                                    |

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                                                       |                                                                                                                                                                     |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Waste treatment methods</u><br>Waste from Residues/Unused Products | Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations |
| Contaminated Packaging                                                | Dispose of this container to hazardous or special waste collection point.                                                                                           |
| Other Information                                                     | Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains                                         |

## SECTION 14: TRANSPORT INFORMATION

|                                |                                                            |
|--------------------------------|------------------------------------------------------------|
| <u>IMDG/IMO</u>                |                                                            |
| UN-No                          | UN1479                                                     |
| Hazard Class                   | 5.1                                                        |
| Packing Group                  | II                                                         |
| Proper Shipping Name           | Oxidizing solid, n.o.s. (TETRAMETHYLAMMONIUM PERCHLORATE)  |
| <u>Road and Rail Transport</u> |                                                            |
| UN-No                          | UN1479                                                     |
| Hazard Class                   | 5.1                                                        |
| Packing Group                  | II                                                         |
| Proper Shipping Name           | Oxidizing solid, n.o.s. (TETRAMETHYLAMMONIUM PERCHLORATE)  |
| <u>IATA</u>                    |                                                            |
| UN-No                          | UN1479                                                     |
| Hazard Class                   | 5.1                                                        |
| Packing Group                  | II                                                         |
| Proper Shipping Name           | OXIDIZING SOLID, N.O.S.* (TETRAMETHYLAMMONIUM PERCHLORATE) |
| Special Precautions for User   | No special precautions required                            |

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## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component                                    | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL |
|----------------------------------------------|-----------|------|-----|-------|------|------|-------|------|------|
| Methanaminium, N,N,N-trimethyl-, perchlorate | 219-805-5 | X    | -   | -     | X    | X    | -     | -    | -    |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date** 22-Mar-2025  
**Revision Summary** Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

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## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**